Supplementary Material for "Free and Sovereign LLMs for Saudi Smart Cities"

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TABLE I
KSA-URBANBENCH KNOWLEDGE ACCURACY BY CITY (%)

Model	Riyadh	Jeddah	Makkah	Madinah	Dammam
ALLaM-7B	**39.0**	25.0	30.0	28.0	28.0
GPT-OSS-20B	38.0	32.0	27.0	29.0	**37.0**
Gemma 4 12B	38.0	**35.0**	**40.0**	**30.0**	26.0
Qwen3-8B	35.0	31.0	36.0	23.0	26.0
Llama 3.1 8B	34.0	27.0	27.0	26.0	26.0

$n = 100$ per city (five knowledge tasks, ten items each, two languages).

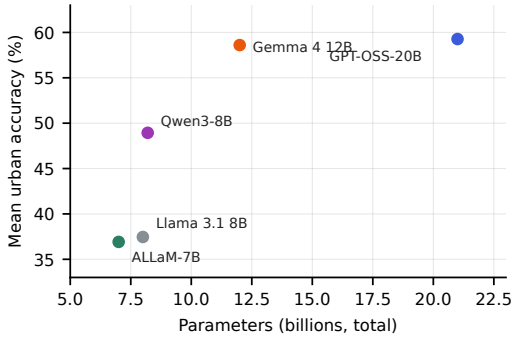


Fig. 1. Mean urban accuracy (STBench+ and KSA-UrbanBench) versus model size.

This supplement reports results summarized in the main paper. All values come from the same evaluation run (NVIDIA A100, November 6–7, 2026); per-item scores and raw outputs are released with the evaluation harness.

I. PER-TASK URBAN ACCURACY

Table II reports accuracy for each of the 16 STBench+ classification tasks.

II. KSA-URBANBENCH BY CITY

Table I reports knowledge accuracy per Saudi city.

III. ACCURACY VERSUS MODEL SIZE

Fig. 1 relates mean urban accuracy to model size.

IV. PAIRWISE SIGNIFICANCE TESTS

Table III lists Holm-corrected p-values for every model pair.

TABLE II
PER-TASK URBAN ACCURACY (%), $n = 50$ PER TASK

	Task	ALLaM-7B	GPT-OSS-20B	Gemma 4 12B	Qwen3-8B	Llama 3.1 8B
KC	POI category recognition	86.0	**98.0**	**98.0**	**98.0**	92.0
KC	POI identification	56.0	90.0	**96.0**	68.0	52.0
KC	Urban region function	34.0	**50.0**	**50.0**	38.0	42.0
KC	Administrative region	54.0	**76.0**	60.0	62.0	66.0
STR	Point–region relation	58.0	**100.0**	**100.0**	**100.0**	74.0
STR	Trajectory–region relation	4.0	**70.0**	48.0	6.0	12.0
STR	Point–trajectory relation	56.0	**100.0**	**100.0**	98.0	36.0
STR	Trajectory identification	46.0	**74.0**	**74.0**	26.0	54.0
AC	Direction determination	16.0	**50.0**	**50.0**	**50.0**	26.0
AC	Trajectory intersection	36.0	**40.0**	**40.0**	**40.0**	30.0
AC	Shortest-path navigation	30.0	**100.0**	**100.0**	98.0	42.0
SR	Rush-hour detection	50.0	80.0	**84.0**	62.0	40.0
SR	Road-level judgment	40.0	68.0	**86.0**	64.0	58.0
SR	Taxi occupancy detection	48.0	64.0	**82.0**	52.0	46.0
DA	Trajectory anomaly detection	60.0	**64.0**	56.0	48.0	48.0
DA	Transport-mode classification	22.0	**74.0**	66.0	40.0	38.0

TABLE III
HOLM-CORRECTED MCNEMAR p-VALUES FOR ALL MODEL PAIRS (BOLD: SIGNIFICANT AT $\alpha = 0.05$)

Pair	Urban	KSA know.	KSA comp.	ArabicMMLU	IFEval	HumanEval+	XSTest
ALLaM–GPT-OSS	**<.001**	1.000	**<.001**	0.337	**<.001**	**<.001**	1.000
ALLaM–Gemma	**<.001**	1.000	**<.001**	**<.001**	**<.001**	**<.001**	**0.027**
ALLaM–Qwen3	**<.001**	1.000	**<.001**	0.050	**<.001**	**<.001**	**<.001**
ALLaM–Llama	0.253	1.000	0.430	**0.018**	**<.001**	**<.001**	1.000
GPT-OSS–Gemma	0.793	1.000	**0.003**	**<.001**	**<.001**	0.774	**0.011**
GPT-OSS–Qwen3	**<.001**	1.000	**<.001**	0.337	0.107	0.237	**<.001**
GPT-OSS–Llama	**<.001**	0.950	**<.001**	**<.001**	0.107	**<.001**	1.000
Gemma–Qwen3	**<.001**	1.000	0.268	**0.001**	0.105	0.191	0.438
Gemma–Llama	**<.001**	0.156	**<.001**	**<.001**	**<.001**	**<.001**	**0.036**
Qwen3–Llama	**<.001**	1.000	**<.001**	**<.001**	**0.001**	**<.001**	**<.001**